Week 2 – pfSense Firewall Configuration & pfBlockerNG Implementation

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Organization : ITSimplera Solutions
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➢ Introduction

AFirewalls play a vital role in protecting modern computer networks by monitoring and controlling incoming and outgoing traffic based on predefined security rules. **pfSense** is a powerful open-source firewall platform that is widely used in enterprise environments due to its reliability, flexibility, and extensive security features.The purpose of this assignment was to gain practical experience in deploying and configuring **pfSense** together with **pfBlockerNG**, an advanced security package that provides **GeoIP filtering**, **DNSBL website blocking**, **IP reputation filtering**, and **threat intelligence integration**.

Throughout this assignment, the following security tasks were successfully completed:

- Installation and configuration of **pfBlockerNG**.
- Implementation of **GeoIP-based country blocking** using the **MaxMind GeoLite2 Database**.
- Configuration of **DNSBL website filtering** to block **Facebook** and **Chat.com**.
- Implementation of **administrator access control** for securing the pfSense WebGUI.
- Validation of firewall rules and monitoring of firewall logs and reports.

This assignment provided valuable practical exposure to firewall administration, network security, and security monitoring while improving my understanding of real-world Security Operations Center (SOC) practices

➢ Executive Summary

This report presents the successful completion of **Week 2** of the **SOC Internship** at **ITSimplera Solutions**. The primary objective of this assignment was to deploy and configure a **pfSense Firewall** and enhance its security capabilities using **pfBlockerNG**.During this assignment, several security features were successfully implemented, including **GeoIP-based country blocking**, **DNSBL website filtering**, **administrator access control**, and **firewall log monitoring**. Each configuration was carefully tested and validated to ensure that the implemented security policies were functioning correctly.This practical exercise provided valuable hands-on experience in enterprise firewall management and improved my understanding of network security, access control, threat prevention, and security monitoring techniques commonly used in a Security Operations Center (SOC).

➢ Objectives

TThe main objectives of this assignment were:

- Successfully deploy the pfSense Firewall.
- Install and configure the pfBlockerNG security package.
- Configure the MaxMind GeoLite2 database for GeoIP filtering.
- Block incoming traffic from selected geographical regions.
- Configure DNSBL to restrict access to selected websites.
- Implement administrator access control for the pfSense WebGUI.
- Validate all security configurations using firewall logs and reports.
- Gain practical experience in enterprise firewall administration.

Learning Outcomes

After successfully completing this assignment, I gained practical knowledge and hands-on experience in the following areas:

- Deploying and configuring pfSense Firewall.
- Installing and managing pfBlockerNG.
- Configuring GeoIP filtering using MaxMind.
- Implementing DNSBL website filtering.
- Creating and managing firewall rules.
- Securing administrative access to network devices.
- Monitoring firewall logs and security reports.
- Validating firewall configurations through practical testing.
- Understanding enterprise firewall management and SOC operations

Lab Environment

The following lab environment was used during the implementation of this assignment.

Component	Details
Host Operating System	Windows 10
Virtualization Software	Oracle VirtualBox
Firewall Platform	pfSense CE
Security Package	pfBlockerNG-devel
DNS Resolver	Unbound DNS
GeoIP Provider	MaxMind GeoLite2
Client Machine	Windows Host
Web Browser	Google Chrome

➢ Software & Hardware Requirements

❖ Software

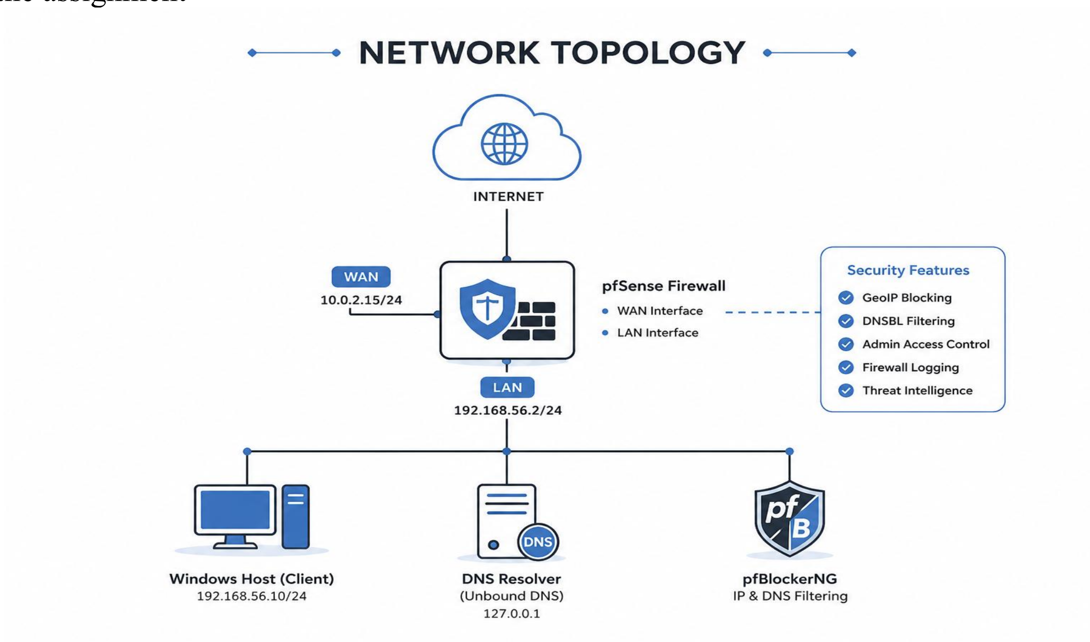
- Oracle VirtualBox
- pfSense CE
- pfBlockerNG-devel
- MaxMind Account
- Windows 10
- Google Chrome

❖ Hardware

- Intel Processor
- 8GB RAM
- 100GB Storage
- Stable Internet Connection

➢ Network Topology

The lab environment consisted of a Windows host machine connected to a pfSense firewall running inside Oracle VirtualBox using a Host-Only Network Adapter. The pfSense firewall acted as the central security gateway, where all firewall rules, GeoIP filtering, DNSBL configurations, and administrator access policies were implemented.The Windows host system was used to access the pfSense WebGUI, perform configuration tasks, and validate the implemented security controls throughout the assignment



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➢ Task 1 – pfSense Firewall Deployment

Objective

The objective of this task was to deploy the pfSense Community Edition firewall in a virtual environment using Oracle VirtualBox and configure its basic network interfaces. The firewall was prepared to serve as the primary security gateway for all subsequent security configurations performed during the internship

Introduction

A firewall is one of the most important security components in any network because it monitors, filters, and controls incoming and outgoing traffic based on predefined security rules. In this lab environment, pfSense was selected because it is a powerful open-source firewall platform widely used by organizations for network security, routing, VPN services, intrusion prevention, and traffic monitoring.

The deployment of pfSense served as the foundation for implementing advanced security features such as GeoIP filtering, DNSBL website blocking, administrator access control, and firewall monitoring using pfBlockerNG.

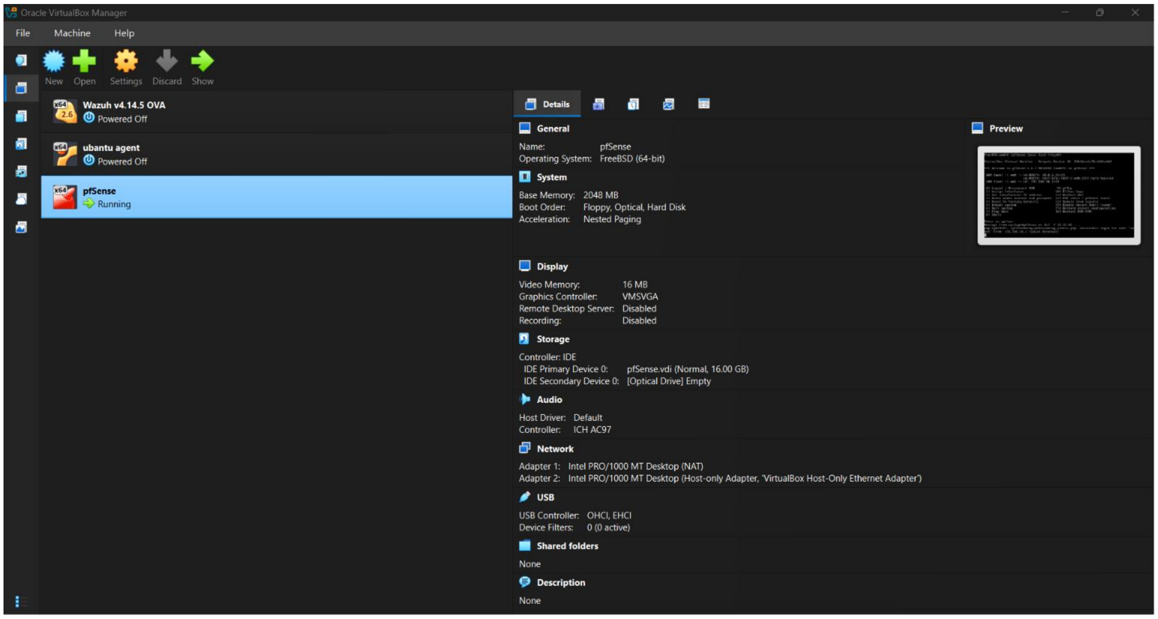
Procedure

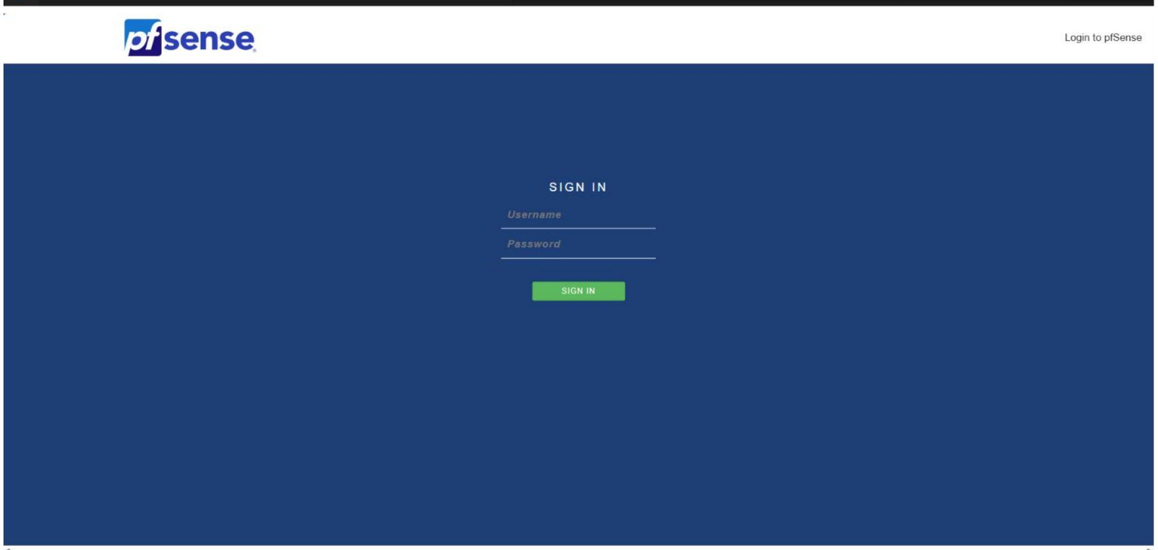
The deployment process began by importing the pfSense virtual appliance into Oracle VirtualBox. During the initial boot process, the WAN and LAN interfaces were assigned according to the lab requirements.

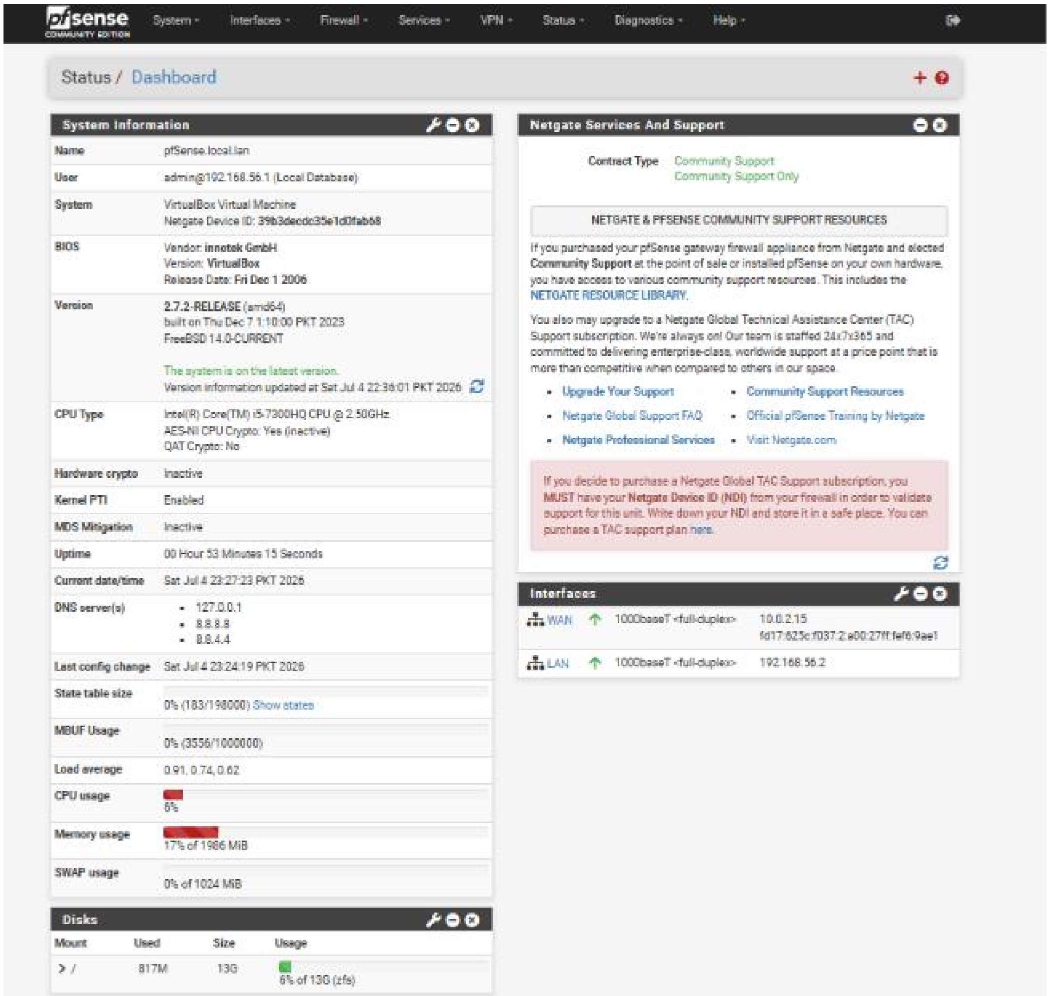
After interface assignment, the LAN interface was configured with the IP address **192.168.56.2/24**, which served as the management interface for accessing the firewall.

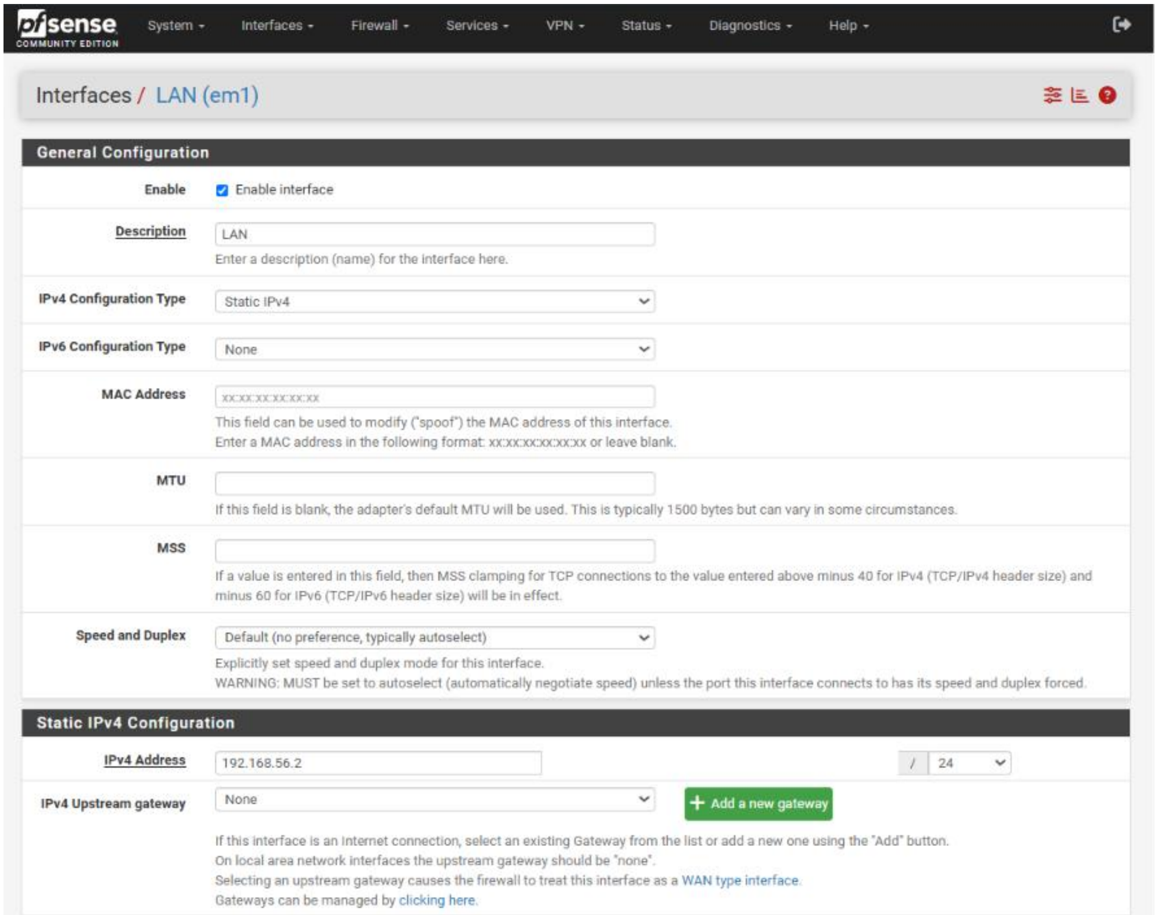
The DHCP server was enabled on the LAN interface so that connected devices could automatically obtain IP addresses. Once the configuration was completed, the Windows host machine successfully communicated with the firewall using ICMP (Ping), confirming proper network connectivity.

Finally, the pfSense WebGUI was accessed through the browser using the configured LAN IP address, where the administrator successfully logged in and verified that the firewall was operational.


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
Wazuh v4.14.5 OVA
Powered Off
ubantu agent
Powered Off
pfSense
Running
Details
General
Name: pfSense
Operating System: FreeBSD (64-bit)
System
Base Memory: 2048 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging
Preview
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Primary Device 0: pfSense.vdi (Normal, 16.00 GB)
IDE Secondary Device 0: [Optical Drive] Empty
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
Adapter 2: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None


pfsense
Login to pfSense
SIGN IN
Username
Password
SIGN IN


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Status / Dashboard
System Information
Name pfSense.local.lan
User admin@192.168.56.1 (Local Database)
System VirtualBox Virtual Machine
Netgate Device ID: 39b3decdc35e1d0fab68
BIOS Vendor: innotek GmbH
Version: VirtualBox
Release Date: Fri Dec 1 2006
Version 2.7.2-RELEASE (amd64)
built on Thu Dec 7 1:10:00 PKT 2023
FreeBSD 14.0-CURRENT
The system is on the latest version.
Version information updated at Sat Jul 4 22:36:01 PKT 2026
CPU Type Intel(R) Core(TM) i5-7300HQ CPU @ 2.50GHz
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto Inactive
Kernel PTI Enabled
MDS Mitigation Inactive
Uptime 00 Hour 53 Minutes 15 Seconds
Current date/time Sat Jul 4 23:27:23 PKT 2026
DNS server(s) 127.0.0.1
8.8.8.8
8.8.4.4
Last config change Sat Jul 4 23:24:19 PKT 2026
State table size 0% (183/198000) Show states
MBUF Usage 0% (3556/1000000)
Load average 0.91, 0.74, 0.62
CPU usage 6%
Memory usage 17% of 1986 MiB
SWAP usage 0% of 1024 MiB
Disks
Mount Used Size Usage
/ 817M 13G 6% of 13G (zfs)
Netgate Services And Support
Contract Type Community Support
Community Support Only
NETGATE & PFSENSE COMMUNITY SUPPORT RESOURCES
If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY.
You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.
Upgrade Your Support
Community Support Resources
Netgate Global Support FAQ
Official pfSense Training by Netgate
Netgate Professional Services
Visit Netgate.com
If you decide to purchase a Netgate Global TAC Support subscription, you MUST have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan here.
Interfaces
WAN 1000baseT <full-duplex> 10.0.2.15
fd17:625c:f037:2:a00:27ff:fef6:9ae1
LAN 1000baseT <full-duplex> 192.168.56.2



Validation

The deployment was verified by successfully accessing the pfSense WebGUI through the Windows host system. Network connectivity was confirmed using the Ping command, and the dashboard displayed both WAN and LAN interfaces in an active state. The successful login confirmed that the firewall was correctly configured and ready for additional security configurations.

Result

The pfSense firewall was successfully deployed and configured within Oracle VirtualBox. The WebGUI became accessible through the configured LAN interface, providing a stable platform for implementing advanced firewall security features during the remaining internship tasks.

Security Benefit

Deploying pfSense provides centralized network security by monitoring, filtering, and controlling all network traffic. It establishes a secure gateway between internal devices and external networks while serving as the primary platform for implementing enterprise-level security policies

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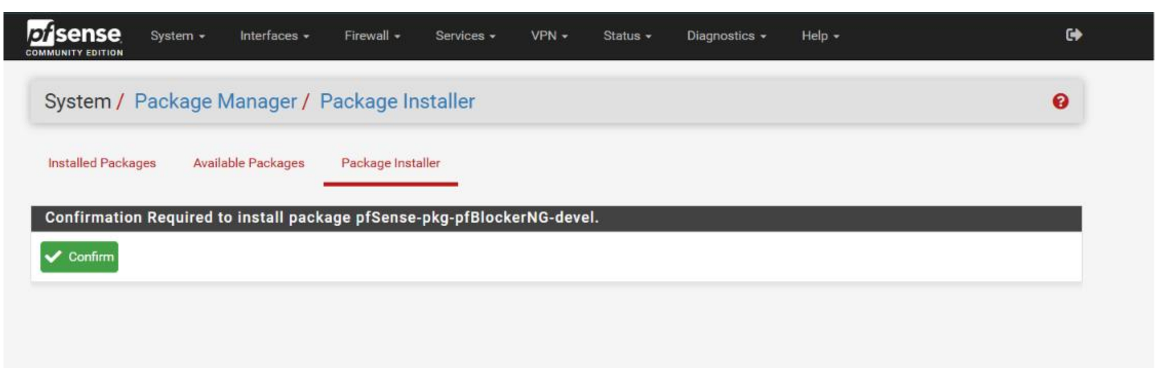
➢ Task 2 – pfBlockerNG Installation

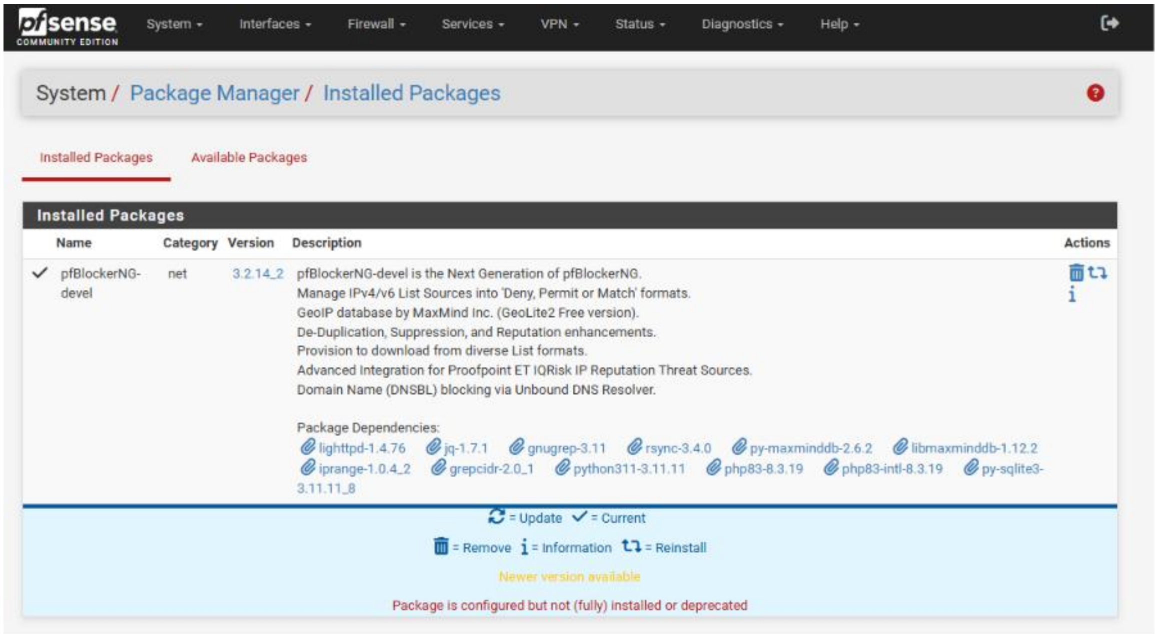
Objective

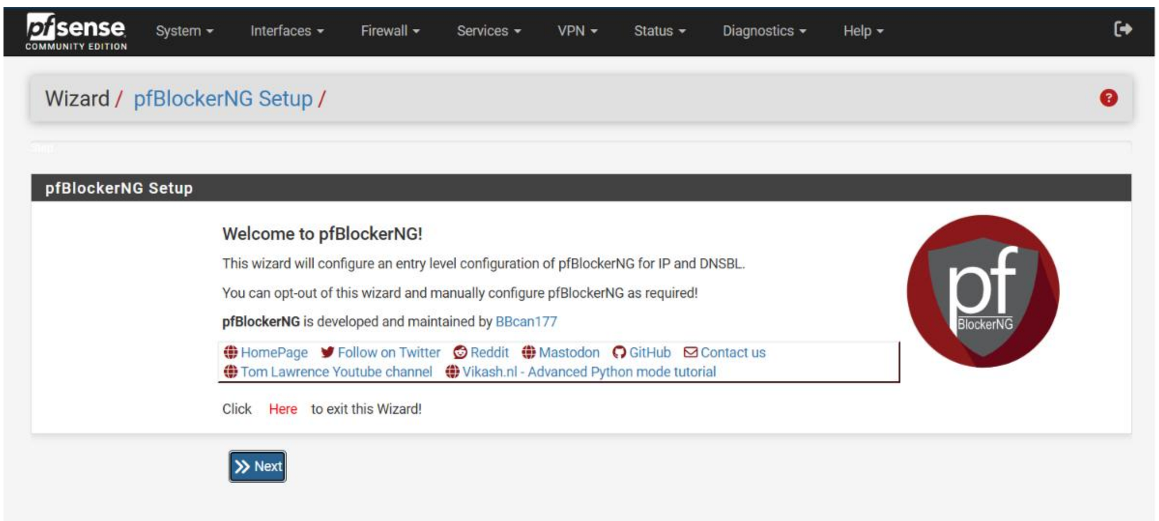
The objective of this task was to install and configure **pfBlockerNG-devel**, an advanced security package available for pfSense Firewall. This package extends the firewall's capabilities by providing IP reputation filtering, GeoIP country blocking, DNSBL (DNS Blackhole List), and threat intelligence integration. Installing pfBlockerNG was an essential step because the remaining security tasks of this assignment depended on its successful deployment

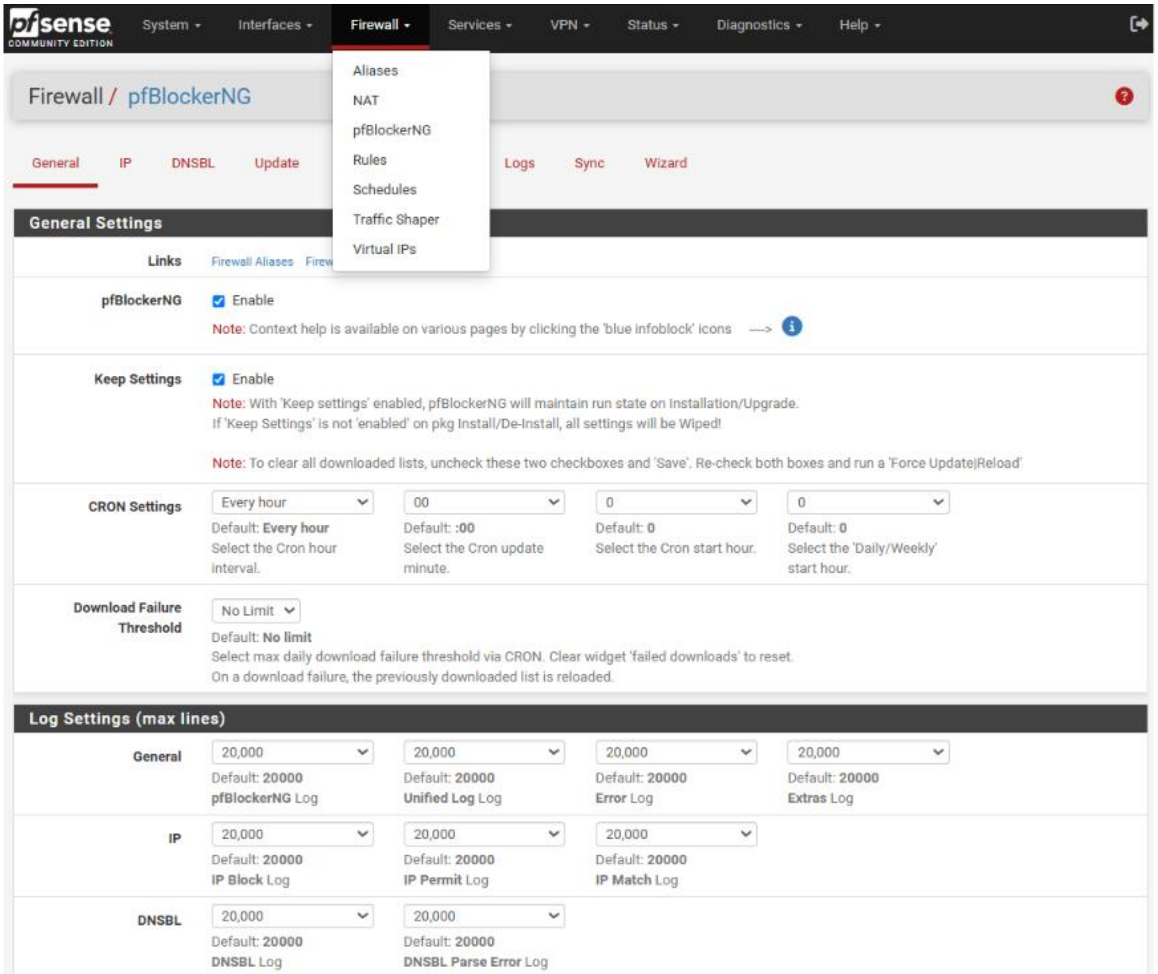
Procedure

The installation process began by accessing the **Package Manager** from the pfSense WebGUI. The Package Manager contains a list of official packages that can be installed to extend pfSense functionality.Using the search feature, the **pfBlockerNG-devel** package was located and selected for installation. After confirming the installation, pfSense automatically downloaded the required files and completed the installation process. Once the installation finished successfully, a new **pfBlockerNG** option appeared under the **Firewall** menu, confirming that the package had been integrated into the firewall. The package was then ready for further configuration, including GeoIP filtering and DNSBL implementation in the subsequent tasks.




sense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
System / Package Manager / Installed Packages
Installed Packages
Available Packages
Installed Packages
Name
Category
Version
Description
Actions
pfBlockerNG-devel
net
3.2.14_2
pfBlockerNG-devel is the Next Generation of pfBlockerNG.
Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats.
GeoIP database by MaxMind Inc. (GeoLite2 Free version).
De-Duplication, Suppression, and Reputation enhancements.
Provision to download from diverse List formats.
Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources.
Domain Name (DNSBL) blocking via Unbound DNS Resolver.
Package Dependencies:
lighttpd-1.4.76
jq-1.7.1
gnugrep-3.11
rsync-3.4.0
py-maxminddb-2.6.2
libmaxminddb-1.12.2
iprange-1.0.4_2
grepcidr-2.0_1
python311-3.11.11
php83-8.3.19
php83-intl-8.3.19
py-sqlite3-3.11.11_8
= Update
= Current
= Remove
= Information
= Reinstall
Newer version available
Package is configured but not (fully) installed or deprecated


sense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Wizard / pfBlockerNG Setup /
pfBlockerNG Setup
Welcome to pfBlockerNG!
This wizard will configure an entry level configuration of pfBlockerNG for IP and DNSBL.
You can opt-out of this wizard and manually configure pfBlockerNG as required!
pfBlockerNG is developed and maintained by BBcan177
HomePage
Follow on Twitter
Reddit
Mastodon
GitHub
Contact us
Tom Lawrence Youtube channel
Vikash.nl - Advanced Python mode tutorial
Click Here to exit this Wizard!
pf
BlockerNG
Next



Validation

The installation was verified by checking the **Installed Packages** section in the Package Manager. Additionally, the appearance of the **pfBlockerNG** option under the **Firewall** menu confirmed that the installation was completed successfully and that the package was ready for configuration.

Result

The **pfBlockerNG-devel** package was installed successfully without any errors. The package became fully operational and was successfully integrated into the pfSense firewall, enabling advanced security features required for the remaining tasks of the internship assignment.

Security Benefit

Installing pfBlockerNG significantly enhances the security capabilities of pfSense by enabling IP reputation filtering, GeoIP-based traffic blocking, DNS-based website filtering, and threat intelligence integration. These features help organizations prevent malicious traffic, reduce exposure to cyber threats, and improve overall network security through centralized firewall management.

➢ Task 3 – GeoIP Country Blocking Using MaxMind

Objective

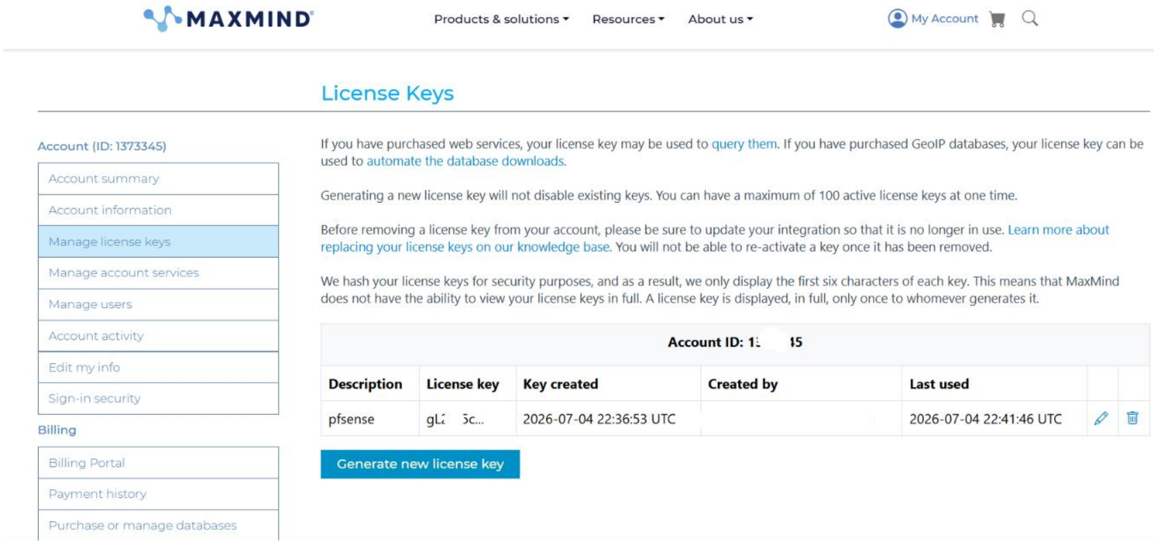
The objective of this task was to configure **GeoIP-based country blocking** using **MaxMind GeoLite2 Database** within pfBlockerNG. This feature was implemented to restrict inbound traffic originating from selected geographical locations, specifically **Netherlands (NL)** and Europe, thereby reducing the risk of unauthorized access and cyber threats.

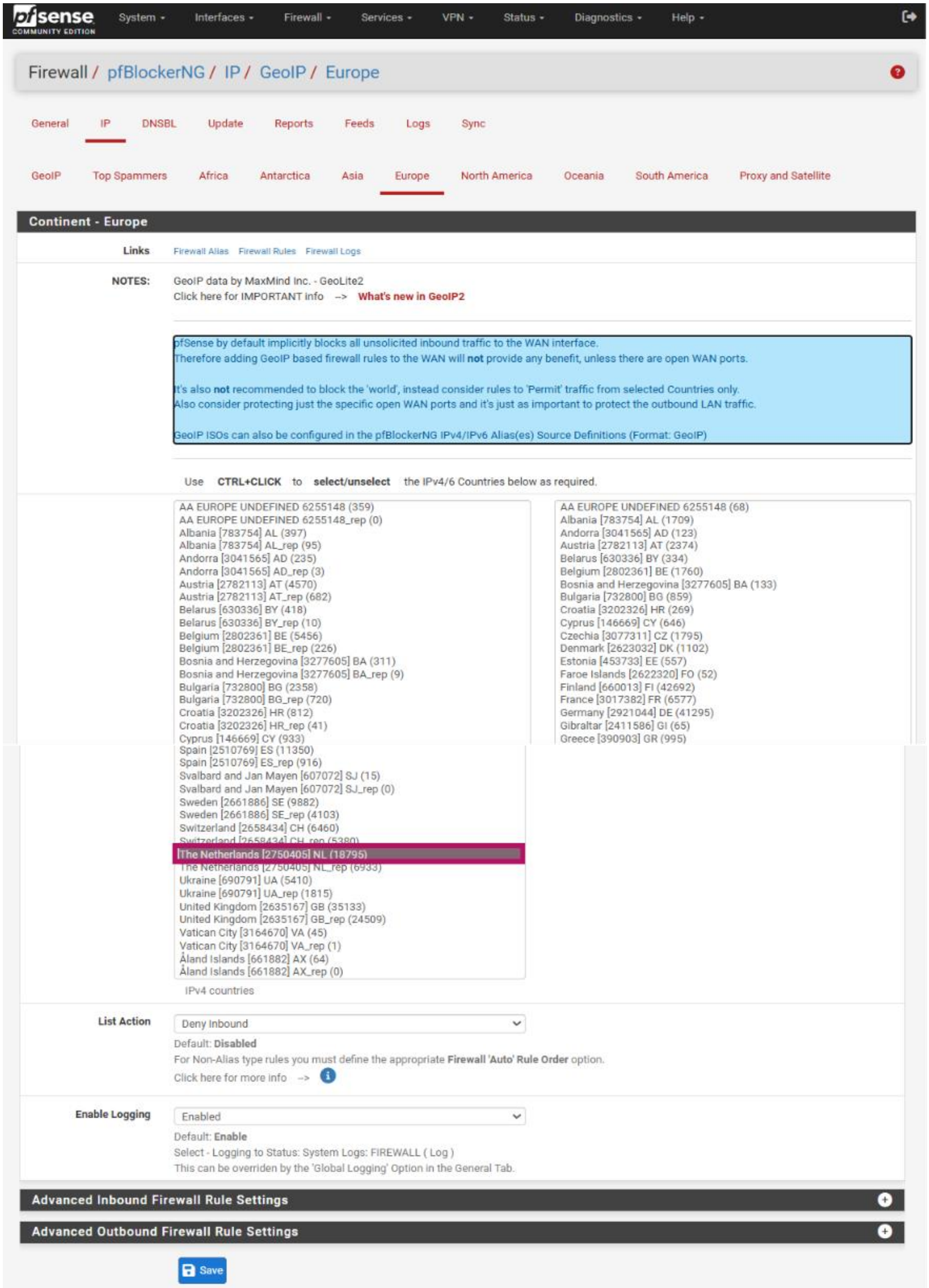
Introduction

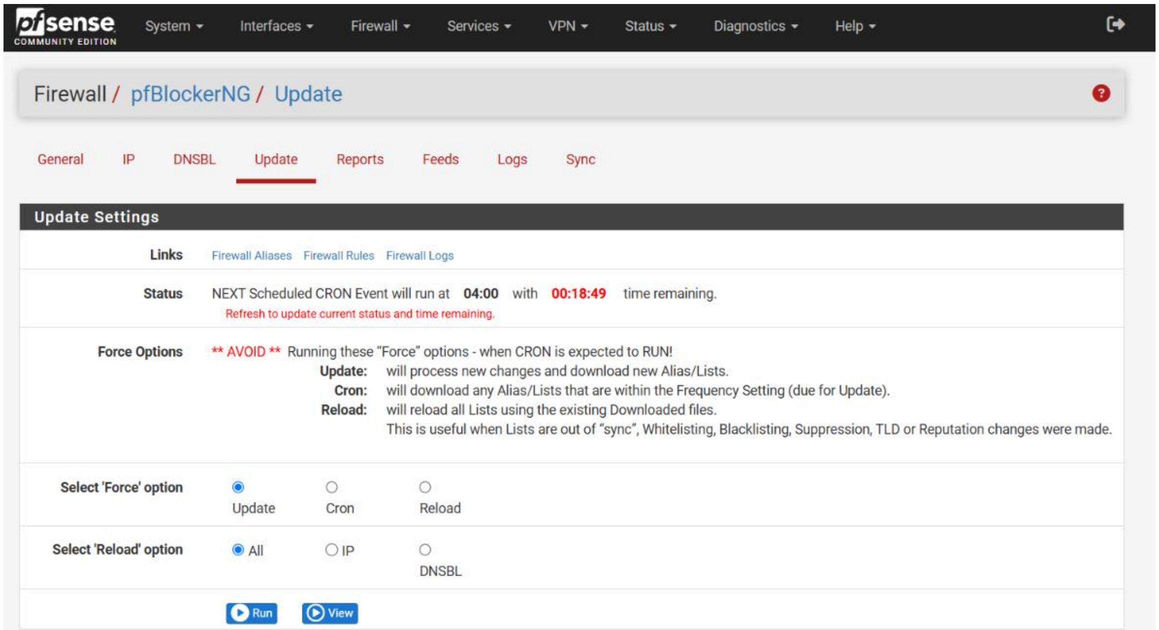
GeoIP filtering is an advanced security feature that allows a firewall to make access control decisions based on the geographical location of an IP address. Instead of allowing traffic from every country, administrators can selectively block or allow connections from specific regions according to organizational security requirements.pfBlockerNG integrates with the **MaxMind GeoLite2** database, which maintains an up-to-date mapping between IP addresses and their geographical locations. During this task, the MaxMind database was connected with pfSense, enabling the firewall to automatically identify IP addresses belonging to the selected countries.To demonstrate this functionality, the **Netherlands (NL)** and the **Europe** region were configured with the **Deny Inbound** action. This ensured that any incoming connection attempts originating from those regions would be blocked before reaching the internal network.

Procedure

The configuration process started by creating a free account on the **MaxMind** website and generating a valid **License Key**. This key was required so that pfBlockerNG could download the latest GeoLite2 database. Inside the pfSense WebGUI, the **Firewall → pfBlockerNG → IP** section was opened. The MaxMind **Account ID** and **License Key** were entered into the GeoIP configuration page and the settings were saved successfully. Next, the **GeoIP** tab was opened, where the **Netherlands (NL)** and the required **Europe** region were selected. The **List Action** was configured as **Deny Inbound**, instructing pfSense to reject all incoming traffic originating from the selected locations. After completing the configuration, the **Force Update / Reload** option was executed. pfBlockerNG downloaded the latest GeoIP database from MaxMind, processed the selected regions, and generated the required firewall aliases. Finally, the generated alias **pfB_Europe_v4** was verified under the Firewall Aliases section, confirming that the selected IP ranges had been successfully loaded into the firewall.

- 
MAXMIND
Products & solutions
Resources
About us
My Account
License Keys
Account (ID: 1373345)
Account summary
Account information
Manage license keys
Manage account services
Manage users
Account activity
Edit my info
Sign-in security
Billing
Billing Portal
Payment history
Purchase or manage databases
If you have purchased web services, your license key may be used to query them. If you have purchased GeoIP databases, your license key can be used to automate the database downloads.
Generating a new license key will not disable existing keys. You can have a maximum of 100 active license keys at one time.
Before removing a license key from your account, please be sure to update your integration so that it is no longer in use. Learn more about replacing your license keys on our knowledge base. You will not be able to re-activate a key once it has been removed.
We hash your license keys for security purposes, and as a result, we only display the first six characters of each key. This means that MaxMind does not have the ability to view your license keys in full. A license key is displayed, in full, only once to whomever generates it.
Account ID:
Description
License key
Key created
Created by
Last used
pfsense
2026-07-04 22:36:53 UTC
2026-07-04 22:41:46 UTC
Generate new license key


pfsense COMMUNITY EDITION
System Interfaces Firewall Services VPN Status Diagnostics Help
Firewall / pfBlockerNG / IP / GeoIP / Europe
General IP DNSBL Update Reports Feeds Logs Sync
GeoIP Top Spammers Africa Antarctica Asia Europe North America Oceania South America Proxy and Satellite
Continent - Europe
Links Firewall Alias Firewall Rules Firewall Logs
NOTES: GeoIP data by MaxMind Inc. - GeoLite2
Click here for IMPORTANT info --> What's new in GeoIP2
pfSense by default implicitly blocks all unsolicited inbound traffic to the WAN interface.
Therefore adding GeoIP based firewall rules to the WAN will not provide any benefit, unless there are open WAN ports.
It's also not recommended to block the 'world', instead consider rules to 'Permit' traffic from selected Countries only.
Also consider protecting just the specific open WAN ports and it's just as important to protect the outbound LAN traffic.
GeoIP ISOs can also be configured in the pfBlockerNG IPv4/IPv6 Alias(es) Source Definitions (Format: GeoIP)
Use CTRL+CLICK to select/unselect the IPv4/6 Countries below as required.
AA EUROPE UNDEFINED 6255148 (359)
AA EUROPE UNDEFINED 6255148_rep (0)
Albania [783754] AL (397)
Albania [783754] AL_rep (95)
Andorra [3041565] AD (235)
Andorra [3041565] AD_rep (3)
Austria [2782113] AT (4570)
Austria [2782113] AT_rep (682)
Belarus [630336] BY (418)
Belarus [630336] BY_rep (10)
Belgium [2802361] BE (5456)
Belgium [2802361] BE_rep (226)
Bosnia and Herzegovina [3277605] BA (311)
Bosnia and Herzegovina [3277605] BA_rep (9)
Bulgaria [732800] BG (2358)
Bulgaria [732800] BG_rep (720)
Croatia [3202326] HR (812)
Croatia [3202326] HR_rep (41)
Cyprus [146669] CY (933)
Spain [2510769] ES (11350)
Spain [2510769] ES_rep (916)
Svalbard and Jan Mayen [607072] SJ (15)
Svalbard and Jan Mayen [607072] SJ_rep (0)
Sweden [2661886] SE (9882)
Sweden [2661886] SE_rep (4103)
Switzerland [2658434] CH (6460)
The Netherlands [2750405] NL (18795)
Ukraine [690791] UA (5410)
Ukraine [690791] UA_rep (1815)
United Kingdom [2635167] GB (35133)
United Kingdom [2635167] GB_rep (24509)
Vatican City [3164670] VA (45)
Vatican City [3164670] VA_rep (1)
Åland Islands [661882] AX (64)
Åland Islands [661882] AX_rep (0)
IPv4 countries
AA EUROPE UNDEFINED 6255148 (68)
Albania [783754] AL (1709)
Andorra [3041565] AD (123)
Austria [2782113] AT (2374)
Belarus [630336] BY (334)
Belgium [2802361] BE (1760)
Bosnia and Herzegovina [3277605] BA (133)
Bulgaria [732800] BG (859)
Croatia [3202326] HR (269)
Cyprus [146669] CY (646)
Czechia [3077311] CZ (1795)
Denmark [2623032] DK (1102)
Estonia [453733] EE (557)
Faroe Islands [2622320] FO (52)
Finland [660013] FI (42692)
France [3017382] FR (6577)
Germany [2921044] DE (41295)
Gibraltar [2411586] GI (65)
Greece [390903] GR (995)
List Action Deny Inbound
Default: Disabled
For Non-Alias type rules you must define the appropriate Firewall 'Auto' Rule Order option.
Click here for more info -->
Enable Logging Enabled
Default: Enable
Select - Logging to Status: System Logs: FIREWALL (Log)
This can be overriden by the 'Global Logging' Option in the General Tab.
Advanced Inbound Firewall Rule Settings
Advanced Outbound Firewall Rule Settings
Save



Validation

The GeoIP configuration was validated after executing the **Force Update / Reload** process. The update completed successfully without errors, and the alias **pfB_Europe_v4** appeared in the Firewall Aliases section containing thousands of IP addresses associated with the selected geographical region.The successful creation of the alias confirmed that pfSense had correctly downloaded the MaxMind GeoLite2 database and applied the GeoIP filtering policy.

Result

The GeoIP filtering policy was successfully implemented using the MaxMind GeoLite2 database. The firewall generated the **pfB_Europe_v4** alias containing the IP ranges associated with the selected European region. As a result, inbound traffic originating from the configured locations was automatically denied, strengthening the overall security of the firewall.

Security Benefit

GeoIP filtering provides an additional layer of defense by allowing administrators to restrict traffic based on geographical regions. This reduces unnecessary exposure to internet-based threats, blocks unwanted connection attempts from selected countries, and helps minimize the organization's attack surface. It also supports organizational security policies by allowing access only from trusted locations while preventing communication from regions considered high risk.

➢ Task 4 – DNSBL Website Filtering

Objective

The objective of this task was to configure **DNSBL (DNS Blackhole List)** using pfBlockerNG to block access to selected websites at the DNS level. In this assignment, **Facebook** and **Chat.com** were added to the custom DNSBL list to demonstrate website filtering and domain blocking within the pfSense firewall environment.

Introduction

DNSBL (DNS Blackhole List) is one of the most powerful features of pfBlockerNG. It works by blocking domain name resolution before a connection to the website is established. Instead of allowing the client device to resolve the real IP address of a blocked website, DNSBL redirects the request to a local **sinkhole IP address**, preventing access to the website.Unlike traditional firewall rules that block traffic after a connection attempt is made, DNSBL stops the communication at the DNS resolution stage. This approach improves network performance while providing an efficient method for blocking malicious, phishing, advertising, or non-work-related websites.During this task, **Facebook** and **Chat.com** were selected as sample domains to verify that DNSBL was functioning correctly.

Procedure

The DNSBL configuration began by creating a **Virtual IP Address**, which acts as the DNSBL sinkhole. This Virtual IP is used to redirect requests for blocked domains instead of allowing them to resolve to their actual public IP addresses.After creating the Virtual IP, the **DNSBL** feature was enabled from the **Firewall → pfBlockerNG → DNSBL** menu.A new DNSBL Group named **SocialBlock** was then created to organize the blocked domains. Inside the **DNSBL Custom List**, the following domains were added:

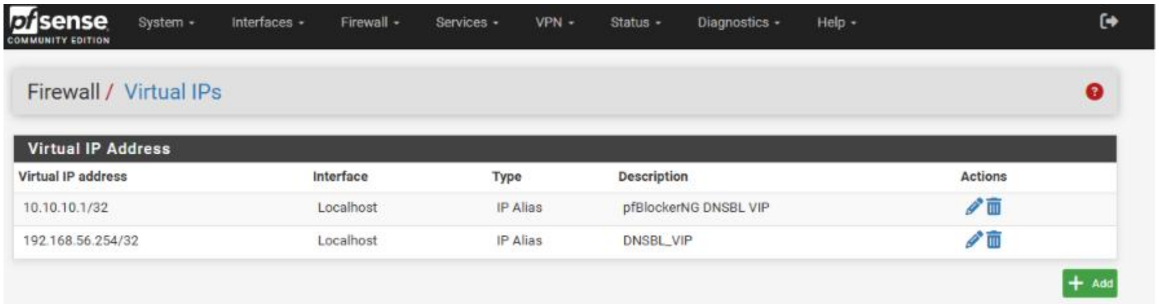
- **facebook.com**
- www.facebook.com
- **chat.com**
- www.chat.com

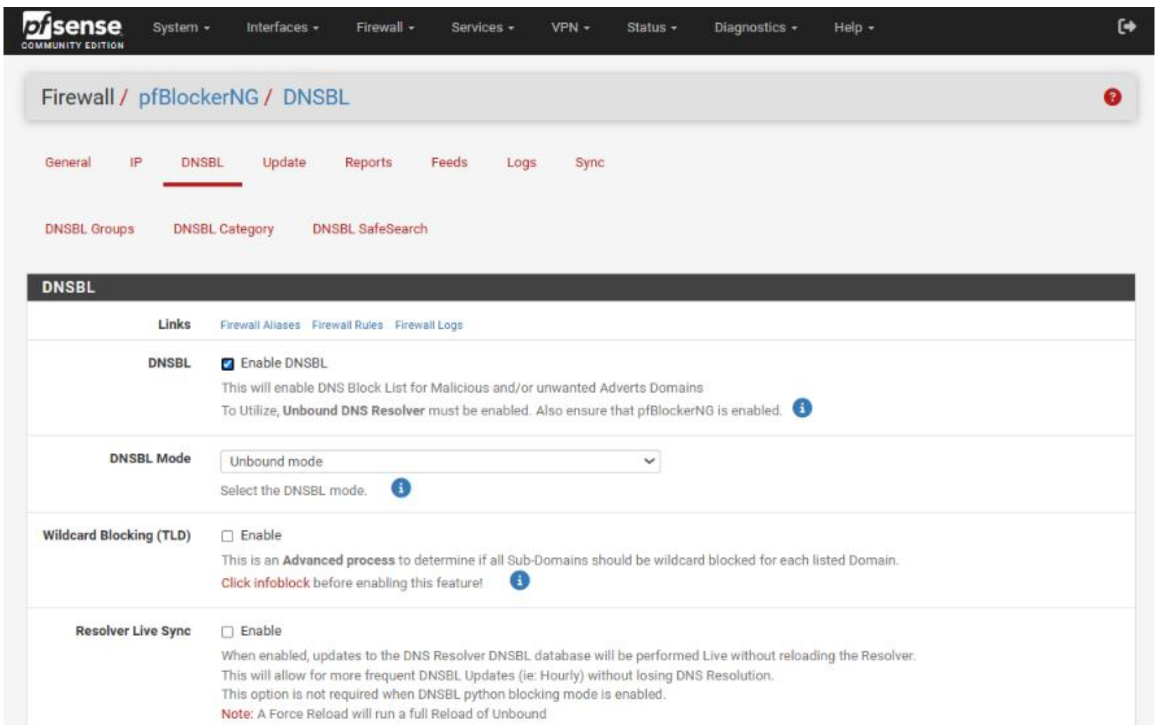
Once the domains were added, the configuration was saved and the **Force Update / Reload** option was executed. pfBlockerNG processed the DNSBL configuration, updated the DNS database, and applied the new filtering rules.Finally, website blocking was verified by attempting to access the blocked domains from the client machine. The browser was unable to establish a connection to the configured websites, confirming that the DNSBL policy was working successfully.

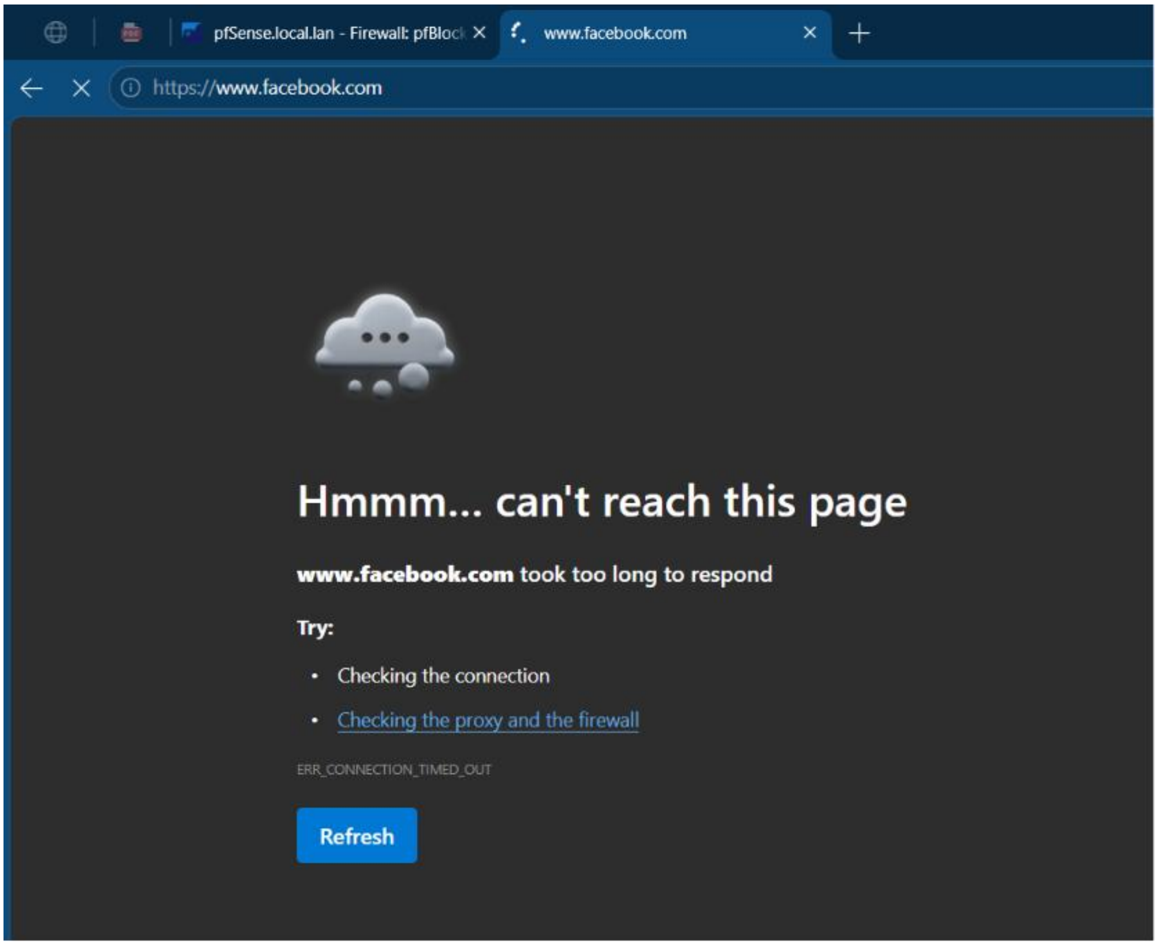
Blocked Domains

The following domains were configured in the DNSBL Custom List:

- facebook.com
- www.facebook.com
- chat.com
- www.chat.com







Validation

The DNSBL configuration was validated by performing practical testing from the Windows host system.After the DNSBL update completed successfully, an attempt was made to open **Facebook** using a web browser. The website failed to load, confirming that the configured DNSBL policy was successfully preventing access to the blocked domain.Additionally, the DNSBL reports within pfBlockerNG were reviewed to verify that the blocked domain requests were being recorded. The successful appearance of these entries confirmed that the DNS filtering policy was functioning correctly.

Result

The DNSBL configuration was successfully implemented using pfBlockerNG. Both **Facebook** and **Chat.com** were added to the custom block list and became inaccessible from the client system. The successful browser test and DNSBL report entries confirmed that the website filtering policy had been applied correctly.

Security Benefit

DNSBL significantly enhances organizational security by preventing users from accessing unauthorized, malicious, or distracting websites before any network connection is established. This reduces the risk of phishing attacks, malware infections, and unnecessary internet usage while helping organizations enforce acceptable use policies. By blocking domains at the DNS level, DNSBL provides an efficient and lightweight method of web filtering without requiring additional software on client machines.

➢ Task 5 – Administrator Access Control

Objective

The objective of this task was to secure the **pfSense WebGUI** by restricting administrative access to only an authorized host. This was achieved by creating a dedicated firewall rule that allowed HTTPS access exclusively from the Windows administrator machine while preventing unauthorized systems on the network from accessing the firewall management interface.

Introduction

The pfSense WebGUI is the primary interface used by network administrators to configure and manage the firewall. Since it provides complete control over firewall settings, protecting access to this interface is a critical security requirement.Allowing unrestricted access to the WebGUI increases the risk of unauthorized configuration changes and potential security breaches. To minimize this risk, only trusted administrator devices should be granted permission to access the firewall management portal.In this task, an **Alias** was created for the administrator's Windows machine, and a firewall rule was implemented to allow **HTTPS (Port 443)** access only from that specific host.

Procedure

The configuration process began by navigating to **Firewall** → **Aliases** in the pfSense WebGUI. A new Host Alias named **Admin_Host** was created, and the IP address of the Windows administrator system (**192.168.56.1**) was assigned to it. This alias provides a simple and organized way to reference the administrator's device in firewall rules.After creating the alias, the **Firewall** → **Rules** → **LAN** section was opened. A new firewall rule was added at the top of the rule list to ensure it would be evaluated before other rules.

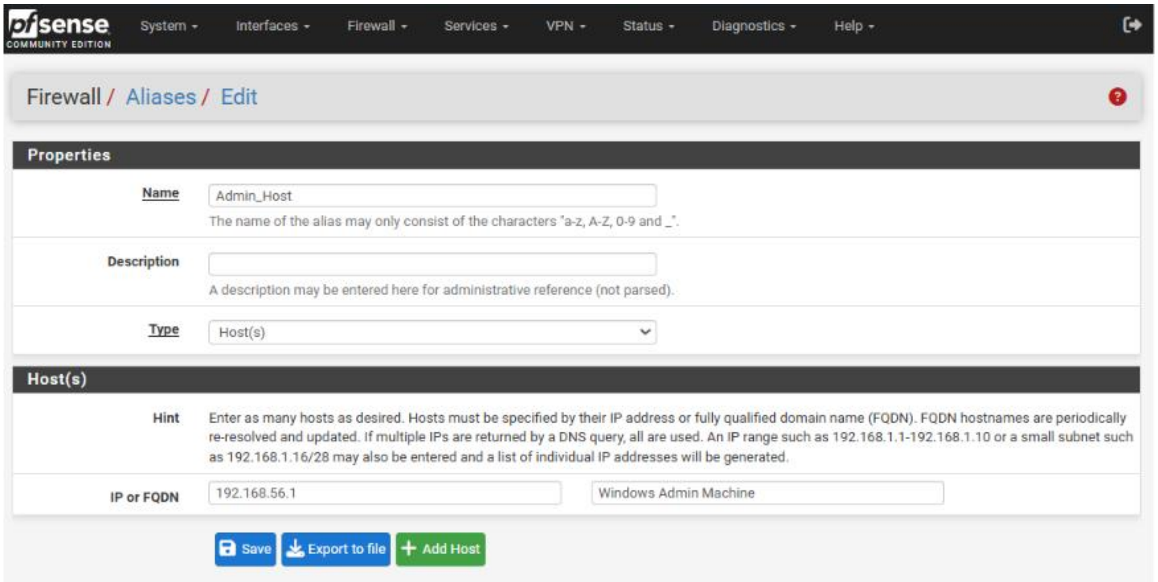
The rule was configured with the following settings:

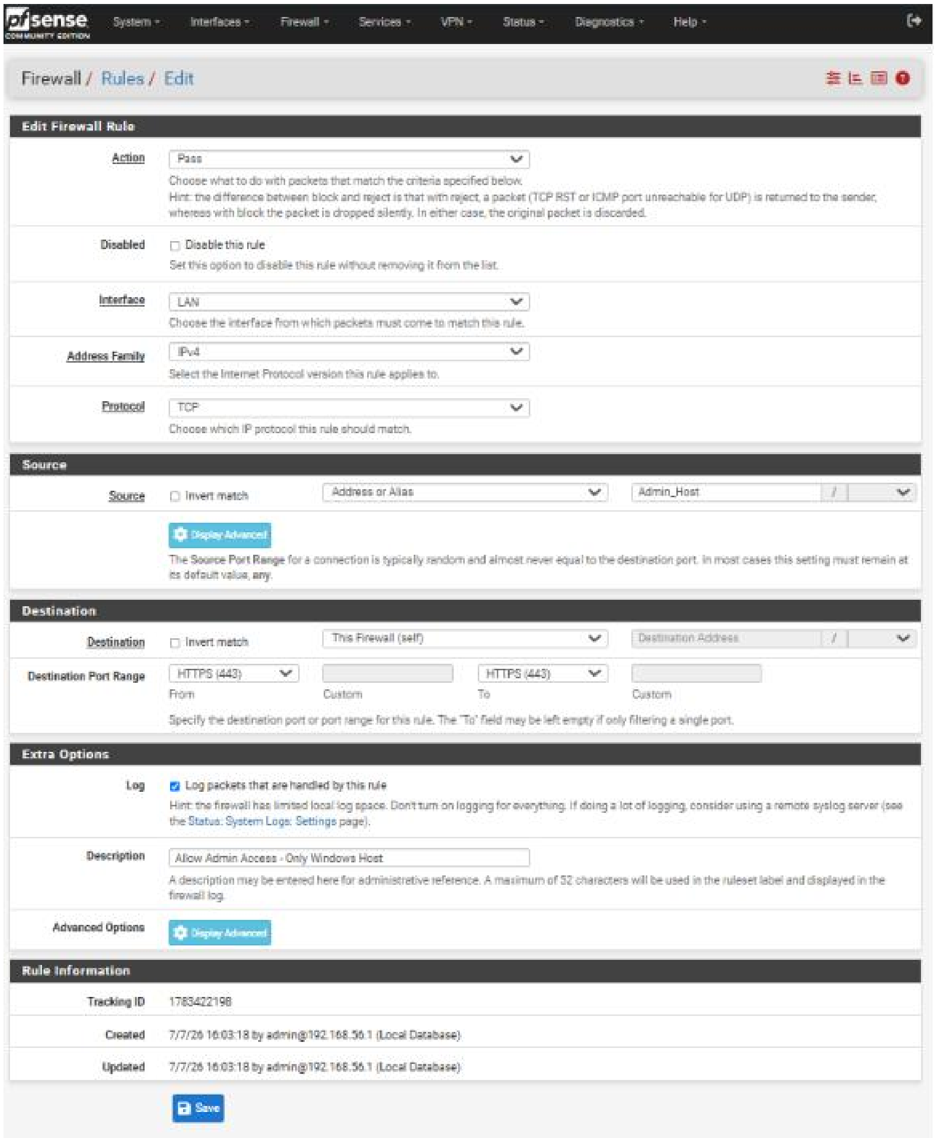
- **Action:** Pass
- **Protocol:** TCP
- **Source:** Admin_Host (192.168.56.1)
- **Destination:** This Firewall (Self)
- **Destination Port:** HTTPS (443)
- **Logging:** Enabled
- **Description:** Allow Admin Access – Only Windows Host

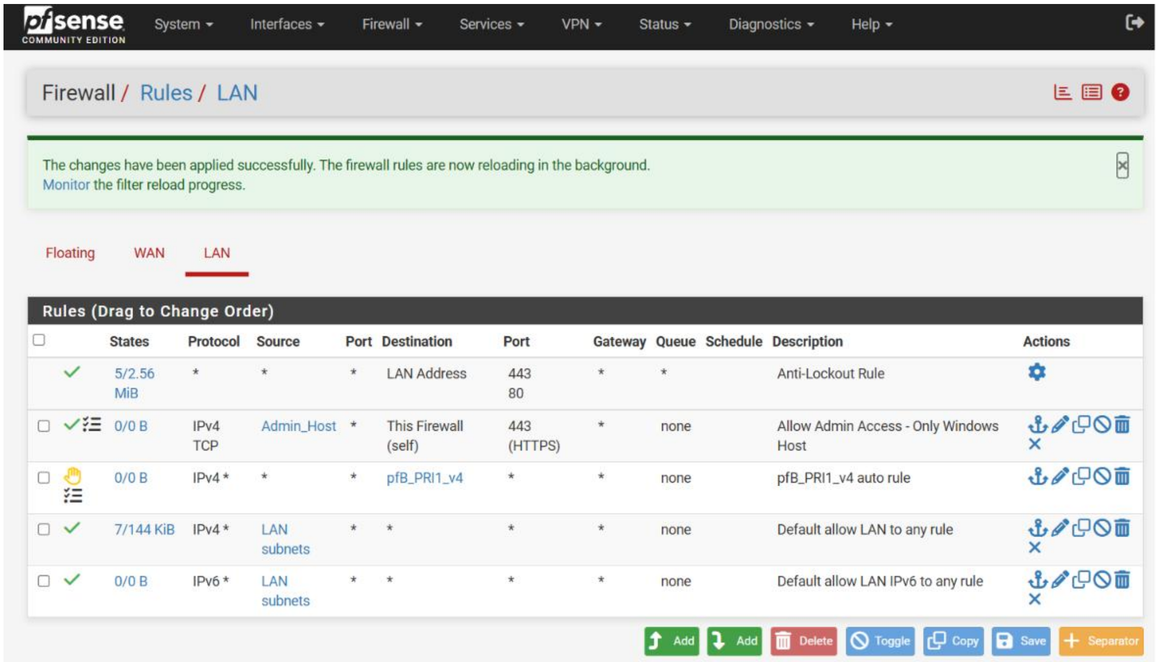
After reviewing the configuration, the rule was saved and **Apply Changes** was selected. The firewall immediately began enforcing the new access control policy.

Firewall Rule Configuration

Setting	Value
Action	Pass
Protocol	TCP
Source	Admin_Host
Destination	This Firewall (Self)
Port	HTTPS (443)
Logging	Enabled


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Firewall / Aliases / Edit
Properties
Name
Admin_Host
The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".
Description
A description may be entered here for administrative reference (not parsed).
Type
Host(s)
Host(s)
Hint
Enter as many hosts as desired. Hosts must be specified by their IP address or fully qualified domain name (FQDN). FQDN hostnames are periodically re-resolved and updated. If multiple IPs are returned by a DNS query, all are used. An IP range such as 192.168.1.1-192.168.1.10 or a small subnet such as 192.168.1.16/28 may also be entered and a list of individual IP addresses will be generated.
IP or FQDN
192.168.56.1
Windows Admin Machine
Save
Export to file
Add Host


System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
LAN
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
Address or Alias
Admin_Host
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
This Firewall (self)
Destination Address
Destination Port Range
HTTPS (443)
HTTPS (443)
From
Custom
To
Custom
Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.
Extra Options
Log
Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).
Description
Allow Admin Access - Only Windows Host
A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.
Advanced Options
Display Advanced
Rule Information
Tracking ID
1783422198
Created
7/7/26 16:03:18 by admin@192.168.56.1 (Local Database)
Updated
7/7/26 16:03:18 by admin@192.168.56.1 (Local Database)
Save



RValidation

The administrator access policy was verified by accessing the pfSense WebGUI from the authorized Windows host machine. The login page loaded successfully, confirming that the firewall rule was functioning as intended.The firewall rule was also reviewed under the **LAN Rules** section, where the configured source, destination, and HTTPS port settings matched the intended security policy. This confirmed that only the designated administrator system had permission to access the firewall management interface.

Result

The administrator access control policy was successfully implemented. Access to the pfSense WebGUI was limited to the authorized Windows host through the configured firewall rule. This ensured that firewall management could only be performed from a trusted device within the network.

Security Benefit

Restricting administrative access is a fundamental security practice in enterprise networks. By allowing only authorized systems to access the pfSense WebGUI, the risk of unauthorized configuration changes, credential misuse, and management interface attacks is significantly reduced. This implementation strengthens the overall security posture of the firewall and ensures that administrative privileges remain protected.

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➢ Task 6 – Firewall Logs & Reports Validation

Objective

The objective of this task was to validate and monitor all previously configured security policies by reviewing the **pfSense** and **pfBlockerNG** firewall logs and reports. This step ensured that the implemented configurations, including **GeoIP blocking**, **DNSBL website filtering**, and **administrator access control**, were functioning correctly and generating the expected security events.

Introduction

Monitoring and log analysis are fundamental responsibilities of a **Security Operations Center (SOC)** analyst. Configuring security controls alone is not sufficient; administrators must continuously verify that these controls are operating correctly and are effectively protecting the network.pfSense provides detailed firewall logs and reporting features that allow administrators to monitor blocked connections, DNSBL events, GeoIP filtering activities, and firewall rule enforcement. These logs help identify suspicious activities, troubleshoot configuration issues, and verify that security policies are working as intended.In this task, the configured security features were validated by reviewing pfBlockerNG reports, checking firewall aliases, and performing practical tests from the Windows host system.

Procedure

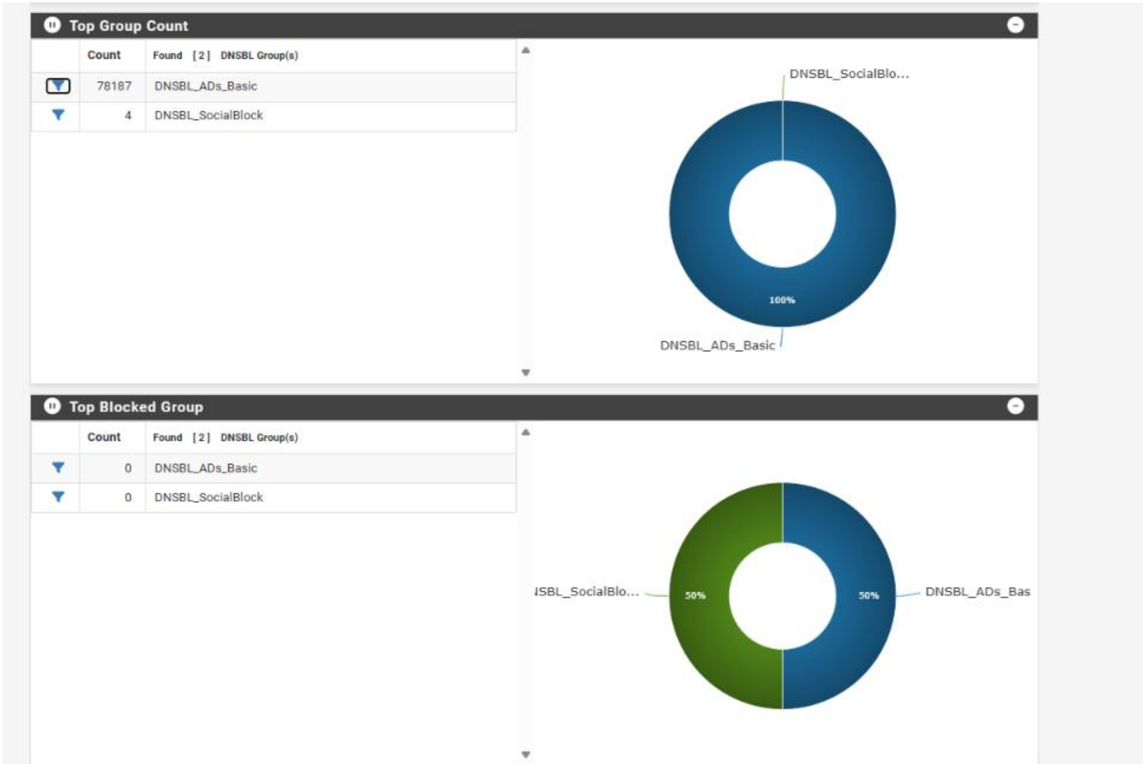
After completing all previous configurations, the **Firewall → pfBlockerNG → Reports** section was opened to review the generated security logs and statistics.The **IP Block Statistics** report was examined to verify that the configured GeoIP rules were active and that the firewall had successfully loaded the blocked IP ranges from the MaxMind GeoLite2 database.Next, the **DNSBL Reports** were reviewed to confirm that the blocked domains, including **Facebook** and **Chat.com**, were being processed correctly by the DNSBL engine.To validate website filtering, an attempt was made to access **Facebook** through a web browser. The website failed to load, confirming that the DNSBL policy was successfully blocking access.The **Firewall Aliases** section was also reviewed to verify that the **pfB_Europe_v4** alias had been created successfully and contained the imported IP addresses associated with the selected geographical regions.Finally, the overall firewall dashboard and reports were reviewed to ensure that all implemented configurations were functioning correctly without any errors.

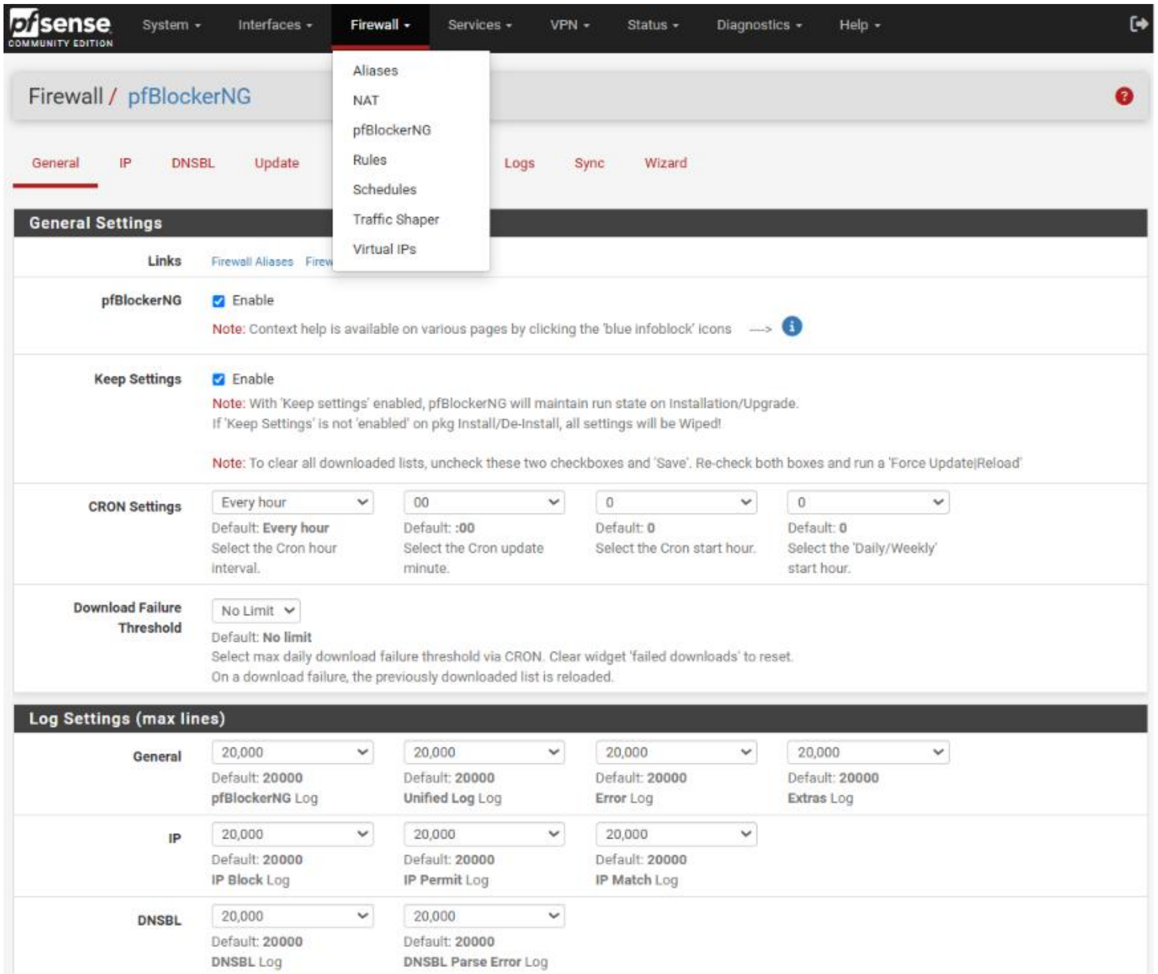
Validation Activities

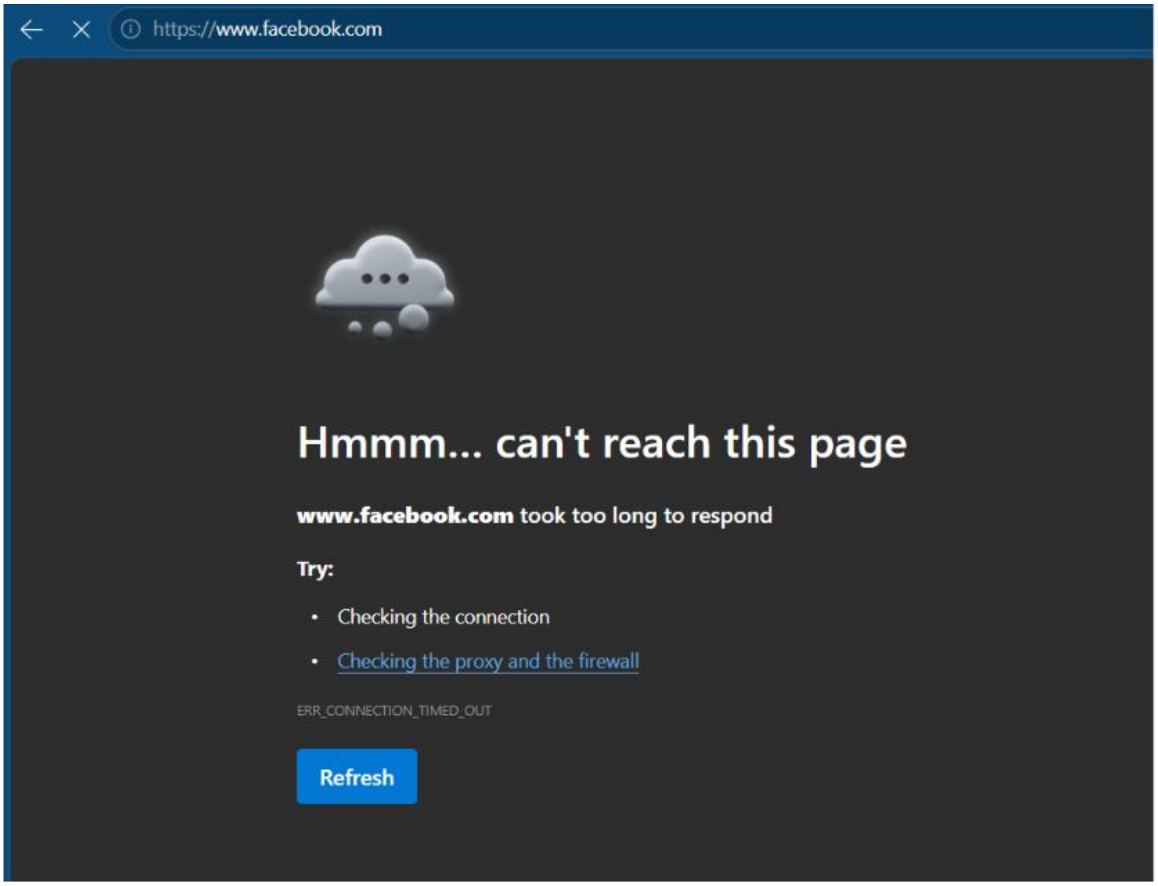
The following validation checks were performed during this task:

- Verified that the **pfB_Europe_v4** alias was successfully created.
- Reviewed **IP Block Statistics** within pfBlockerNG.
- Checked **DNSBL Reports** for blocked domain entries.

- Confirmed that **Facebook** and **Chat.com** were inaccessible through the web browser.
- Reviewed firewall logs to ensure that security events were being recorded successfully.
- Confirmed that all implemented security configurations were operating without errors.




sense
COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Aliases
NAT
pfBlockerNG
Rules
Schedules
Traffic Shaper
Virtual IPs
Firewall / pfBlockerNG
General
IP
DNSBL
Update
Logs
Sync
Wizard
General Settings
Links
Firewall Aliases
pfBlockerNG
Enable
Note: Context help is available on various pages by clicking the 'blue infoblock' icons
Keep Settings
Enable
Note: With 'Keep settings' enabled, pfBlockerNG will maintain run state on Installation/Upgrade.
If 'Keep Settings' is not 'enabled' on pkg Install/De-Install, all settings will be Wiped!
Note: To clear all downloaded lists, uncheck these two checkboxes and 'Save'. Re-check both boxes and run a 'Force Update|Reload'
CRON Settings
Every hour
Default: Every hour
Select the Cron hour interval.
00
Default: :00
Select the Cron update minute.
0
Default: 0
Select the Cron start hour.
0
Default: 0
Select the 'Daily/Weekly' start hour.
Download Failure Threshold
No Limit
Default: No limit
Select max daily download failure threshold via CRON. Clear widget 'failed downloads' to reset.
On a download failure, the previously downloaded list is reloaded.
Log Settings (max lines)
General
20,000
Default: 20000
pfBlockerNG Log
Unified Log Log
Error Log
Extras Log
IP
IP Block Log
IP Permit Log
IP Match Log
DNSBL
DNSBL Log
DNSBL Parse Error Log



Result

The validation process confirmed that all firewall security policies were successfully implemented and operating as expected. GeoIP filtering effectively loaded and enforced the configured country-blocking rules, while DNSBL successfully prevented access to the specified websites. The firewall logs and reports accurately recorded security events, providing clear visibility into network activity and confirming the successful deployment of all required security controls.

Security Benefit

Firewall logs and reports provide continuous visibility into network activity and are essential for detecting, investigating, and responding to security incidents. Regular log monitoring enables administrators to identify suspicious behavior, verify the effectiveness of implemented security policies, and maintain compliance with organizational security standards. By validating the configured firewall rules and DNSBL policies, the overall reliability and effectiveness of the network security infrastructure were successfully confirmed.

➢ Project Summary

After completing all six tasks, the pfSense firewall was successfully configured with **pfBlockerNG**, enabling enterprise-grade security features such as **GeoIP-based country blocking**, **DNSBL website filtering**, **administrator access control**, and **continuous firewall monitoring**. The implementation demonstrated practical knowledge of firewall administration, threat prevention, and SOC operations while providing hands-on experience with tools commonly used in real-world cybersecurity environments

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➢ Challenges Faced

Introduction

During the implementation of this assignment, several technical and configuration-related challenges were encountered. These issues are common while deploying a firewall in a virtual lab environment. Each challenge was analyzed carefully and resolved through troubleshooting, documentation review, and repeated testing.

❖ Challenge 1 – Virtual Machine Network Connectivity

Initially, the Windows host system was unable to communicate with the pfSense firewall because the network adapter was not configured correctly. After verifying the Host-Only Adapter settings in Oracle VirtualBox and reconfiguring the LAN interface, connectivity was successfully established.

❖ Challenge 2 – pfBlockerNG Installation

During the installation process, downloading the package required a stable internet connection. After verifying internet connectivity and refreshing the Package Manager, the installation completed successfully.

❖ Challenge 3 – GeoIP Database Configuration

The MaxMind GeoLite2 database required a valid Account ID and License Key before downloading the GeoIP database. Once the credentials were configured correctly, the database was updated successfully.

❖ Challenge 4 – DNSBL Configuration

The DNSBL configuration required proper Virtual IP settings and a Force Update before website filtering became active. After updating the DNSBL database, the blocked websites became inaccessible as expected.

❖ Challenge 5 – Firewall Rule Verification

Careful attention was required while creating firewall rules to ensure that the administrator access policy was correctly applied. After reviewing the firewall rule order and configuration, the desired access restrictions were successfully implemented.

➢ Results

The implementation of all assigned tasks was completed successfully. Each security feature was configured, tested, and verified using pfSense reports, browser testing, and firewall logs.

Task	Status
pfSense Deployment	✓ Successfully Completed
pfBlockerNG Installation	✓ Successfully Completed
GeoIP Blocking	✓ Successfully Completed
DNSBL Website Filtering	✓ Successfully Completed
Administrator Access Control	✓ Successfully Completed
Firewall Logs & Reports Validation	✓ Successfully Completed

❖ Overall Outcome

The firewall successfully blocked the configured geographical locations using GeoIP filtering, prevented access to Facebook and Chat.com through DNSBL, restricted administrator access to the authorized system, and generated firewall logs confirming that all security policies were functioning correctly.

➢ Skills Gained

This internship assignment provided valuable hands-on experience with enterprise firewall administration and security monitoring. The following technical skills were developed during the completion of this task:

- pfSense Firewall Deployment and Configuration
- VirtualBox Network Configuration
- pfBlockerNG Installation and Management
- GeoIP Filtering using MaxMind
- DNSBL Website Filtering
- Firewall Rule Configuration
- Access Control Implementation
- Security Log Monitoring
- Network Troubleshooting
- Technical Documentation Writing
- SOC Workflow Understanding
- Cybersecurity Best Practices

➢ Conclusion

The successful completion of this assignment provided practical exposure to enterprise firewall technologies and real-world Security Operations Center (SOC) activities. Throughout the implementation process, pfSense and pfBlockerNG were configured to strengthen network security by applying multiple layers of protection, including GeoIP country blocking, DNSBL website filtering, administrator access control, and continuous firewall monitoring.

Each configuration was tested and validated through firewall logs, browser verification, and security reports, confirming that all implemented security controls were functioning correctly. This assignment significantly enhanced my understanding of firewall management, network defense strategies, and security monitoring techniques while improving my practical cybersecurity skills and technical documentation abilities.

➢ References

ITSimplera Solutions – SOC Internship Training Materials:

1. **Netgate.** *pfSense Documentation.* https://docs.netgate.com/pfsense/en/latest/
2. **pfBlockerNG Documentation.** https://www.pfblockerng.com/
3. **MaxMind GeoLite2 Documentation.** https://www.maxmind.com/
4. **Oracle VirtualBox User Manual.** https://www.virtualbox.org/wiki/Documentation

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